

HDOC Day-37 — String Operations in C

Aim

To write a **menu-driven C program** using separate functions to:

1. Count spaces, digits, and special characters in a string.
2. Count the frequency of a given character.
3. Toggle the case of each character.
4. Exit.

Algorithm

1. Start the program.
2. Read a string from the user using `fgets()`.
3. Display the menu.
4. Read the user's choice.
5. If choice is **1**, call `countCharacters()`:
 - o Traverse the string.
 - o Count spaces, digits, and special characters.
6. If choice is **2**, ask for a character and call `frequency()`:
 - o Traverse the string.
 - o Count how many times the given character occurs.
7. If choice is **3**, call `toggleCase()`:
 - o Convert uppercase letters to lowercase.
 - o Convert lowercase letters to uppercase.
 - o Leave digits, spaces, and special characters unchanged.
8. If choice is **4**, terminate the program.
9. For any other choice, display "Invalid choice!".
10. Repeat until the user selects Exit.

C Program

```
#include <stdio.h>

// Function to count spaces, digits and special characters
void countCharacters(char str[])
{
    int spaces = 0, digits = 0, special = 0;
    int i;

    for (i = 0; str[i] != '\0'; i++)
    {
        if (str[i] == ' ')
            spaces++;

        else if (str[i] >= '0' && str[i] <= '9')
            digits++;
    }
}
```

```

        else if (!((str[i] >= 'A' && str[i] <= 'Z') ||
                    (str[i] >= 'a' && str[i] <= 'z')) &&
                    str[i] != '\n')
            special++;
    }

    printf("Number of spaces = %d\n", spaces);
    printf("Number of digits = %d\n", digits);
    printf("Number of special characters = %d\n", special);
}

// Function to count frequency of a character
void frequency(char str[], char ch)
{
    int count = 0;
    int i;

    for (i = 0; str[i] != '\0'; i++)
    {
        if (str[i] == ch)
            count++;
    }

    printf("Frequency of '%c' = %d\n", ch, count);
}

// Function to toggle case
void toggleCase(char str[])
{
    int i;

    for (i = 0; str[i] != '\0'; i++)
    {
        if (str[i] >= 'A' && str[i] <= 'Z')
            str[i] = str[i] + 32;

        else if (str[i] >= 'a' && str[i] <= 'z')
            str[i] = str[i] - 32;
    }

    printf("String after toggling case: %s", str);
}

int main()
{
    char str[200];
    char ch;
    int choice;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    do
    {
        printf("\n----- MENU ----- \n");
        printf("1. Count spaces, digits and special characters\n");

```

```

printf("2. Count frequency of a character\n");
printf("3. Toggle case of each character\n");
printf("4. Exit\n");

printf("Enter your choice: ");
scanf("%d", &choice);

switch (choice)
{
    case 1:
        countCharacters(str);
        break;

    case 2:
        printf("Enter character to find frequency: ");
        scanf(" %c", &ch);
        frequency(str, ch);
        break;

    case 3:
        toggleCase(str);
        break;

    case 4:
        printf("Program terminated.\n");
        break;

    default:
        printf("Invalid choice!\n");
}

} while (choice != 4);

return 0;
}

```

Test Cases

Test Case	Input String	Operation	Expected Result
1	Hello 123!	Count characters	Spaces = 1, Digits = 3, Special = 1
2	banana	Frequency of a	Frequency = 3
3	Hello World123!	Toggle case	hELLO wORLD123!
4	C@Program 2026#	Count characters	Spaces = 1, Digits = 4, Special = 2
5	Programming	Frequency of m	Frequency = 2

Test Case Simulation

Test Case 1

```
→ ./a.out
Enter a string: Hello 123!

—— MENU ——
1. Count spaces, digits and special characters
2. Count frequency of a character
3. Toggle case of each character
4. Exit
Enter your choice: 1
Number of spaces = 1
Number of digits = 3
Number of special characters = 1
```

Test Case 2

```
→ ./a.out
Enter a string: banana

—— MENU ——
1. Count spaces, digits and special characters
2. Count frequency of a character
3. Toggle case of each character
4. Exit
Enter your choice: 2
Enter character to find frequency: a
Frequency of 'a' = 3
```

Test Case 3

```
→ ./a.out
Enter a string: Hello World123!

—— MENU ——
1. Count spaces, digits and special characters
2. Count frequency of a character
3. Toggle case of each character
4. Exit
Enter your choice: 3
String after toggling case: hELLO wORLD123!
```

Test Case Evaluation

The program correctly identifies **spaces, digits, and special characters** while traversing the string. The newline inserted by `fgets()` is excluded from the special-character count.

The frequency function compares every character of the string with the user-provided character and correctly calculates its number of occurrences.

The toggle function changes every **uppercase letter to lowercase and every lowercase letter to uppercase**, while digits, spaces, and special characters remain unchanged.

Result

Thus, a menu-driven C program using **separate functions** was successfully implemented to perform character counting, frequency counting, and case toggling operations on a string.